

National University of Computer and Emerging Sciences, Lahore Campus

Course:	Programming Fundamentals	Course Code:	CS1002	Marks:
Program:	BS (SE)	Semester:	Fall-2025	
Duration:	30 Minutes	Total Marks:	15	Name:
Quiz Date:		Weight		Roll No:
Section:	BSE-1A	Page(s):	2	Quiz-2 (A)

Question 1: Predict the Output (5 Marks)

Analyze the following C++ program. Write down the exact output it will produce.
If the program contains any error, identify it clearly.

Program Segment	Output
<pre>#include <iostream> using namespace std; int main() { int arr[5] = {2, 4, 6, 8, 10}; for (int i = 1; i < 5; i++) { if (arr[i] % 4 == 0) { arr[i - 1] += arr[i] / 2; } else if (arr[i] % 3 == 0) { arr[i] = arr[i - 1] + 3; } else { arr[i]--; } } for (int i = 0; i < 5; i++) cout << arr[i] << " "; return 0; }</pre>	4 4 8 9 (4)

Question 2: Write C++ Code (10 Marks)

Write a C++ program to check whether a number is a "Recursive Divisor Number".

A number n is a *Recursive Divisor Number* if:

- The sum of all its proper divisors (excluding itself) gives another number s , and
- The sum of divisors of s (again excluding itself) equals the original number n .

$n = 220 \rightarrow$ divisors = $1 + 2 + 4 + 5 + 10 + 11 + 20 + 22 + 44 + 55 + 110 = 284$

$284 \rightarrow$ divisors = $1 + 2 + 4 + 71 + 142 = 220$

- `int sumOfDivisors(int n)` $\rightarrow$ uses a **loop** to return the sum of all proper divisors of n .
- `bool isRecursiveDivisor(int n)` $\rightarrow$ calls `sumOfDivisors()` twice and checks the relationship.

Sample run:

Input: Enter a number: 220

Output: 220 is a Recursive Divisor Number.

```
#include <iostream>
using namespace std;
int sum-of-divisors (int n) {
    int sum = 0;
    for (int i = 1; i < n; i++) {
        if (n % i == 0)
            sum += i;
```

(Q.3)

```
    }
    for (int j = sum; j <= sum; j++) {
        sum-divisor = 0;
        if (sum % j == 0)
            sum-divisor += j;
```

```
    }
    if (sum == sum-divisor)
        cout << n << "is a recursive number";
    else
        cout << n << "is not a recursive number";
```

```
int main () {
    int n;
    cout << "Enter a number : ";
    cin >> n;
    sum-of-divisors (int n);
    return 0;
}
```

function ?